Team Name

SMART INDIA HACKATHON 2026

TITLE PAGE

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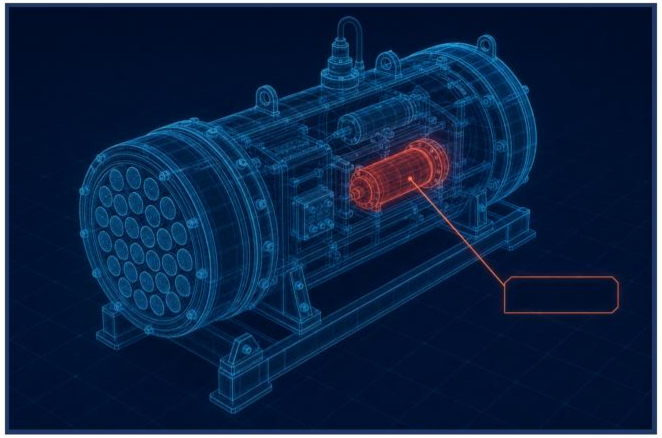
Problem Statement ID	26075
Problem Statement Title	CAPACITY CONNECT — A Digital Capacity Building and Learning Management Portal
Theme	Smart Education
PS Category	Software
Team ID	Team ID
Team Name (Registered on portal)	Team Name



Digital Twin concept (illustrative)

ORGANISATION

Ministry of Earth Sciences (MoES)

DEPARTMENT

India Meteorological Department

One portal for training, assessment and competency — that keeps working when the network does not.

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IDEA TITLE

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CAPACITY CONNECT — one offline-first portal for learning, assessment, competency mapping and AI-assisted field troubleshooting

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- **Single portal, three roles. Trainee, Trainer and Admin — secure signup/login with Admin approval and role management.**
- **Trainee: professional profile (qualifications, experience, skills, interests, certificates), course enrollment, learning resources, subject-wise MCQ assessments and course feedback.**
- **Trainer: competency declaration, questionnaires with deadlines, participation and performance monitoring, and a Trainer Library of recorded lectures, presentations and study material.**
- **Admin: user approval, role management, dashboards for courses / enrollments / certifications / assessments / participation, plus homepage publishing of notifications, announcements and achievements.**
- **Competency mapping engine scores trainers against subject requirements to objectively identify the right trainer.**
- **Offline-first edge app: a real local database plus a local AI node on 127.0.0.1, so work continues at sea and at remote sites.**

How it addresses the problem

- Replaces scattered drives, mail and spreadsheets with one auditable source of truth for learning and competency.
- Removes the connectivity barrier: every feature works offline and reconciles automatically on reconnect.
- Turns tacit instrument know-how into retrievable, guided diagnostic procedures.

Concept UI mockup — offline dashboard and mobile learning app

Innovation and uniqueness of the solution

Offline is the default, not a fallback — the local DB is an operating database, not a cache

Local AI engine, no cloud LLM — deterministic retrieval pipeline on 127.0.0.1

Learning fused with operations — a fault maps to the asset and to the training that covers it

3D Digital Twin guidance — SONAR-001 → Mesh_042 shown CRITICAL from a cached GLB

Objective competency mapping — fit computed from skills, certificates and outcomes

Auditable sync ledger — every offline action is versioned and conflict-checked

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TECHNICAL APPROACH

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Technologies to be used (programming languages, frameworks, hardware)

CLIENT

- Flutter 3 / Dart, Material 3
- Riverpod state · GoRouter
- Drift + SQLite (SQLCipher)
- flutter_secure_storage · Dio
- Android · Windows · Linux · Web

LOCAL AI

- **Python 3.11 · FastAPI on 127.0.0.1**
- **Normalize · language detect · tokenize**
- **Keyword / phrase / RapidFuzz match**
- TF-IDF knowledge retrieval
- Pluggable local LLM slot

3D & MEDIA

- GLTF / GLB cached locally
- **Mesh mapping + fault state layer**
- Chunked resumable downloads
- Streaming video · PDF viewer
- Optional speech-to-text input

BACKEND & OPS

- FastAPI · SQLAlchemy · Pydantic
- **PostgreSQL + Alembic migrations**
- Celery workers · JWT + refresh
- Docker · GitHub Actions CI
- pytest · flutter test/analyze

Methodology and process for implementation

A. System architecture — edge application works with zero connectivity; cloud platform is independently deployable

Flutter Client (Material 3) → Local AI Node 127.0.0.1 → Local DB SQLCipher → Digital Twin GLTF/GLB → Sync Ledger (encrypted) → FastAPI /api/v1 → PostgreSQL + workers → Admin Portal & Dashboards

B. Local AI troubleshooting pipeline — returns structured JSON, confidence is a defined weighted score

Input → Normalize → Tokenize → Keyword + Phrase → Fuzzy Match → Knowledge Retrieval → Component + Severity → 3D Mesh Mapping

Worked example: "Sonar transducer showing abnormal vibration and casing fracture" → SONAR-001 · Sonar Transducer Array · Mesh_042 · severity HIGH · confidence 0.94 → Digital Twin highlight + diagnostic procedure + diagnostic record saved offline.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

Technically proven stack

Flutter, FastAPI and PostgreSQL are mature and fully open-source; one Dart codebase covers Android, Windows, Linux and Web.

No exotic hardware

Runs on existing MoES/IMD field tablets and vessel workstations; backend is containerised on existing infrastructure.

Zero licensing cost

Entirely open-source; no per-seat fees, so scaling to more users and institutes costs only storage and compute.

Incremental rollout

Portal features usable from day one; AI and Digital Twin layers extend the same schema without rework.

Potential challenges and risks → Strategies for overcoming these challenges

CHALLENGE / RISK	MITIGATION STRATEGY
Poor / no connectivity at sea and remote sites	Offline-first architecture: local operating DB + local AI node; sync ledger drains automatically on reconnect.
Large 3D models on low-end field devices	Decimated LOD meshes by default, full-resolution GLB as optional download; renderer degrades to metadata-only view.
Offline credential caching weakens authentication	Admin-approved devices only, Argon2id verifier (never the password), bounded grace window, audit + re-verification on sync.
Sync conflicts on shared records	Per-entity strategies (version compare, field merge, manual queue); diagnostic records are never silently overwritten.
Digital literacy and user adoption	Role-tailored simple UI, localisation-ready strings for Indian languages, optional voice input, in-app guided demo scenario.
Storage exhaustion on field devices	Storage manager with per-category usage, resumable downloads, permissioned removal of optional assets only.

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IMPACT AND BENEFITS

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Potential impact on the target audience

TRAINEES / FIELD ENGINEERS

- **Learn, test and certify with zero connectivity — at sea and at remote sites**
- **Guided AI diagnostics instead of waiting for shore support**
- Portable, verifiable competency profile

TRAINERS

- **One versioned library for lectures, presentations and material**
- **Questionnaires with deadlines, auto-scoring, live participation view**
- Feedback shows which content actually works

ADMINISTRATION / MoES

- **Live dashboards of enrollment, certification and assessment**
- **Objective trainer selection via competency mapping**
- **Complete audit trail for governance and reporting**

Benefits of the solution (social, economic, environmental)

SOCIAL

Equitable access to training for remote and shipboard staff; skill recognition through verifiable certificates; localisation-ready for Indian-language expansion; safer field work through guided procedures.

ECONOMIC

Lower instrument downtime and fewer avoidable failures; reduced travel and physical-classroom cost; open-source stack with no licensing fees; reusable across INCOIS, NCPOR and NIOT on the same core.

ENVIRONMENTAL

Fewer service trips and paper-free assessments; longer instrument life through timely preventive maintenance; better-maintained ocean and weather sensors mean higher-quality earth-science data.

100% core features usable offline

↓ 40% target fault-resolution time

3 roles, one governed platform

4 platforms, one codebase

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

Problem & organisational context

- **SIH 2026 Problem Statement 26075 — Ministry of Earth Sciences (MoES) — sih.gov.in**
- moes.gov.in | India Meteorological Department — mausam.imd.gov.in
- MoES Annual Report — capacity building & human-resource development
- National Education Policy 2020 — digital, competency-based learning
- SIH 2026 portal guidelines — idea submission format and evaluation criteria

Standards & frameworks referenced

- SCORM / xAPI (Experience API) — learning-record interoperability
- IEEE 1484 LOM — learning-object metadata for the course catalog
- NIST SP 800-63B — digital identity & authentication guidance
- OWASP ASVS + Mobile Top 10 — application security verification baseline
- NIST SP 800-38D — AES-256-GCM authenticated encryption
- ISO/IEC 27001 Annex A — access control, logging, audit-trail controls

Technology documentation

- Flutter & Dart — docs.flutter.dev | Riverpod, GoRouter, Drift
- FastAPI — fastapi.tiangolo.com | SQLAlchemy, Alembic, Pydantic official docs
- SQLCipher — full-database encryption for SQLite (zetetic.net/sqlcipher)
- Khronos glTF 2.0 specification — 3D asset format for the Digital Twin
- PostgreSQL 16 docs | Docker Compose deployment reference

Domain & prior art studied

- Digital Twin concepts for marine instrumentation (ISO 23247 framing)
- Argo programme documentation — argo.ucsd.edu — float operations
- Offline-first patterns; CRDT vs. ledger-based synchronisation literature
- Open-source LMS study (Moodle, Open edX) — role model & gap analysis
- RapidFuzz / TF-IDF retrieval literature — basis of the confidence score

Note: all cited standards and libraries are open or publicly available; the solution uses no proprietary or licence-restricted component.